

Enhancing Intelligent Recruitment With Generative Pretrained Transformer and Hierarchical Graph Neural Networks: Optimizing Resume-Job Matching With Deep Learning and Graph-Based Modeling

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ABSTRACT

With the rapid rise of online recruitment, automating resume-job matching is crucial for hiring efficiency. However, current methods fail to capture complex semantic relationships, lack hierarchical representations, and overlook bidirectional compatibility. To address these issues, the authors propose a resume-job matching model integrating generative pretrained transformer 4 embeddings, hierarchical graph neural networks, and a bilateral matching algorithm. This approach effectively models semantic, structural, and bidirectional relationships. Experiments on public datasets show that the authors' model achieves an F1 score of 0.91, outperforming cosine similarity (0.70) and SuperGlue (0.88). Statistical analysis validates the improvements, while ablation studies highlight each module's role. Results show that the authors' approach enhances accuracy and reduces bias, providing a data-driven solution for fair and efficient recruitment.

KEYWORDS

Bilateral Matching Algorithm, GPT-4o Embeddings, Hierarchical Graph Neural Networks, Intelligent Recruitment, Resume-Job Matching, Semantic Relationships

INTRODUCTION

With the acceleration of globalization and digitization, enterprises are facing unprecedented challenges in talent recruitment and management. The ability to efficiently and accurately select and recruit the right candidates has become one of the key factors for maintaining competitiveness, driving

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innovation, and ensuring long-term growth (Boiko et al., 2021; Kurek et al., 2024). Importantly, recruitment is not simply about filling vacancies; in fact, it represents a critical task that directly influences a company's overall performance and its future direction. Talent is one of the most valuable resources for any organization, and the quality of recruitment has a direct impact on organizational performance and success (Maree & Shehada, 2024). As the demand for specialized skills and innovation capabilities grows, the recruitment process has become increasingly complex. Companies must not only match candidates to current job openings but also consider the evolving nature of future roles, expanding responsibilities, and cultural fit. Traditional recruitment methods, such as manual resume screening and interview evaluations, no longer suffice to meet these complex needs (Kambur & Yildirim, 2023; Piccolo et al., 2024). At the same time, advancements in technology—particularly breakthroughs in natural language processing (NLP) and deep learning—offer powerful opportunities to optimize recruitment systems.

Recruitment processes have long relied on human experience and rule-based systems. Typically led by human resources departments, these methods involve manually screening resumes, conducting interviews, and assessing candidates (Haryan et al., 2024). What should be noticed is that these approaches present significant limitations, particularly in terms of efficiency and accuracy. Resume screening often depends on keyword matching and subjective judgments, both of which can miss important qualifications or strengths in candidates' profiles (Maggu & Bhambri, 2024). Interviews, although useful for evaluating soft skills and cultural fit, are time-consuming and subjective, especially when dealing with large numbers of candidates. Additionally, traditional recruitment methods struggle to keep pace with rapidly changing job requirements, resulting in lengthy recruitment cycles and lower accuracy. These shortcomings make traditional approaches less effective, particularly for large-scale or complex hiring needs. With advances in NLP technologies and the success of generative pretrained transformers (GPTs) in language understanding and generation tasks, recruitment systems are increasingly moving toward more intelligent, automated solutions (Drushchak, 2024). GPT-based recruitment systems can analyze vast amounts of data, such as resumes, cover letters, and interview transcripts, to extract deeper insights into candidates' abilities, backgrounds, and fitness for specific roles (Fang et al., 2023; Kolesnikov & Golovko, 2023). These systems go beyond simple keyword matching by understanding candidates' experiences, skills, potential, and cultural compatibility, providing a more precise and comprehensive analysis of fit. GPT-based recruitment systems can automate key aspects of the recruitment process, including resume screening, interview evaluation, and match optimization. For instance, these systems can extract key information from resumes, filter candidates based on job requirements, and assess their fit with job descriptions using natural language generation techniques (Devaraju, 2022; Glazko et al., 2024). By leveraging deep learning algorithms, the systems dynamically adjust candidate selection criteria based on shifts in market trends and business needs, thereby improving both efficiency and accuracy (Skondras et al., 2023). Furthermore, GPT-based systems excel at contextual understanding and information extraction, helping reduce human bias in decision-making and enhancing fairness and transparency in the process (Tsoutsanis & Tsoutsanis, 2024).

Despite the significant advancements offered by GPT-based automated recruitment systems, several challenges remain in managing complex recruitment scenarios and modeling multidimensional information. For example, traditional methods often fail to account for the dynamic nature of candidates' career paths and the hierarchical structure of job positions; this limitation makes it difficult for current recruitment systems to perform deep, multi-angle matching (Paavola, 2024). Thus, combining GPT's deep semantic understanding with structured information modeling remains a crucial area for improvement.

Basing our research on these aforementioned issues and the current situation, we propose an intelligent recruitment model. This method combines the OpenAI GPT-4 omni (GPT-4o) pretrained language model, hierarchical graph neural network (hierarchical GNN), and bilateral matching algorithm. This method not only leverages the powerful deep semantic understanding capabilities of

GPT-4 but also models the multidimensional information in recruitment scenarios (such as candidates' career paths and skill relationships, as well as the hierarchical structure of positions). This effort helps enhance the model's adaptability to complex recruitment scenarios. In addition, the introduction of the bilateral matching algorithm ensures optimization from both the job seekers' and employers' perspectives. In this way, the model's accuracy and flexibility in matching candidates to positions witness great improvement.

The contributions of this study are reflected in these three aspects:

- We propose a method that combines deep semantic understanding with structured information modeling. By integrating the GPT-4o language model with a hierarchical GNN, we designed a model that can capture both the deep semantic relationships and structured matching relationships between candidates and positions, overcoming the limitations of traditional methods that rely solely on keyword matching or simple rules.
- Our model dynamically optimizes the matching process by considering both the career backgrounds of job seekers and the specific requirements of job positions. This capability avoids the one-sidedness of traditional systems that rely on unidirectional evaluation, thus improving the accuracy and fairness of matching.
- We built a recruitment model with adaptive learning capabilities. Through deep learning technology, our model continuously learns and optimizes from historical recruitment data. It adjusts screening criteria in real time according to changes in corporate recruitment needs, demonstrating high adaptability and scalability.

Overall, our model not only improves recruitment efficiency and accuracy but also provides a scalable solution for the intelligent recruitment field. Additionally, it offers new ideas for enterprises to optimize talent management processes in dynamic market environments. This research lays a solid foundation for the design and application of future intelligent recruitment systems.

RELATED WORK

Applications of NLP in Recruitment

NLP, as an important branch of artificial intelligence, focuses on studying the interaction between computers and human language, enabling machines to understand, analyze, generate, and respond to natural language (Fan et al., 2023; Paavola, 2024). With the continuous advancement of NLP technology, especially breakthroughs in deep learning models, its application in the recruitment field has made significant progress, greatly enhancing the efficiency and accuracy of recruitment processes. In the recruitment process, tasks such as resume screening, cover letter analysis, and job description generation involve a large amount of textual data. Traditional recruitment methods primarily rely on manual evaluations, where recruiters have to manually review hundreds or even thousands of resumes; this process is not only time-consuming but also prone to bias (Alamelu et al., 2021). The introduction of NLP technology provides an intelligent solution to this problem.

Resume screening is one of the key steps in the recruitment process. Candidates use their resumes to showcase their educations, work experiences, and skills, which recruiters manually review to assess their suitability for a position. Nevertheless, the formats, language styles, wording, and expression methods of resumes vary greatly, making it easy for manual screening to overlook potential talents or be influenced by subjective bias (Kaygin, 2023; Subramanian et al., 2024; Zhenhua, 2024). In this context, NLP technology, by analyzing keywords, semantic structures, and contextual relationships in resumes, can automatically identify and extract key information about candidates. As such, it can help determine their compatibility with the position (Appadoo et al., 2020; Bhor et al., 2021; Wang, Li, & He, 2025). For example, NLP-based systems

can identify technical skills, job titles, years of experience, and other relevant details, matching them to specific job criteria.

In addition to resume analysis, NLP technology can also be applied to cover letter evaluation (Campion et al., 2024; Wenting et al., 2024). Cover letters typically express a candidate's motivation and interest in a specific position, and recruiters use them to further screen suitable candidates (Harsha et al., 2022). However, the content of cover letters is often lengthy and subjective, making manual analysis challenging. NLP technology can automate the processing of cover letters using sentiment analysis and semantic understanding techniques to extract key information, identify candidates' attitudes, interests, and compatibility (Pudasaini et al., 2022). This technology not only reduces labor costs but also evaluates candidates' potential attitudes and motivations through sentiment analysis.

In the interview evaluation stage, NLP technology is also beginning to play a significant role. Traditional interview evaluations often rely on the subjective judgment of interviewers, who may apply different evaluation standards owing to individual differences (Chen, 2024; Senarathne et al., 2021; Sharma et al., 2021). NLP can analyze the language used during interviews to identify characteristics such as candidates' communication skills, problem-solving abilities, and emotional stability (Sinha et al., 2021). Using speech recognition technology, NLP technology can convert spoken content from interviews into text and then analyze it using sentiment analysis, semantic understanding, and other NLP techniques to automatically evaluate candidates' performances during interviews, providing recruiters with more objective references.

NLP can also be applied to the optimization and generation of job descriptions. Job descriptions are a critical part of the recruitment process, and well-crafted descriptions can attract more suitable candidates. However, writing job descriptions often requires balancing the amount of information and appeal; this task can be a challenge for recruiters. Through natural language generation technology, NLP can automatically generate job descriptions, intelligently recommending suitable wording and formats based on company needs and job requirements (Alsaif et al., 2022; Sannidhan et al., 2023; Soni et al., 2020). This process ensures that job descriptions not only accurately convey job demands but also possess sufficient attractiveness.

In summary, the application of NLP technology in the recruitment field now spans multiple aspects, including resume screening, cover letter analysis, interview evaluation, and job description generation. By increasing the level of automation in the recruitment process, NLP not only optimizes recruitment workflows and reduces human intervention but also plays a critical role in improving the accuracy and efficiency of recruitment (Kang et al., 2020). With continuous technological advancements, NLP will demonstrate even greater potential in the recruitment field, driving the recruitment process toward greater intelligence and personalization.

Applications of Algorithm Optimization in Recruitment

Algorithm optimization has become a core tool for improving the efficiency and accuracy of recruitment systems. Existing studies have shown that algorithm optimization is widely applied in various stages of recruitment, such as resume screening, candidate matching, and job recommendation (Barmuta & Grishchenko, 2020; Gao et al., 2022; Witt & Burke, 2002). Common algorithms include supervised learning, unsupervised learning, deep learning, reinforcement learning, and recommendation system algorithms, all of which play a critical role in advancing the intelligence of recruitment processes. In resume screening, supervised learning algorithms are commonly used to classify candidates based on their characteristics, such as work experience, educational background, and skills. Algorithms like support vector machines, decision trees, and random forests have proven highly effective in this context. For instance, support vector machine maximizes the margin between classes, which enables the model to accurately determine whether a candidate meets the job requirements (Delecraz et al., 2022; Hao et al., 2024). Decision trees and random forests, by constructing a series of decision rules, can categorize candidates into different groups, helping recruitment systems identify suitable candidates from a large pool of resumes (Gao et al., 2022;

Kaushik et al., 2023; Zhang et al., 2025). The advantage of these algorithms lies in their efficiency in processing structured data, particularly when resume data have clear labels and fixed formats, yielding good results. However, their limitations lie in their weaker ability to handle complex textual data, such as cover letters and interview records. To address these limitations, deep learning technologies have been increasingly applied to recruitment systems in recent years. Deep learning algorithms such as deep neural networks, convolutional neural networks, and recurrent neural networks (RNNs) show great potential in automatically extracting features from data, especially in handling unstructured data, such as text, speech, and images (Allal-Chérif et al., 2021; Kodipalli et al., 2023; Wang, Li, Lv, et al., 2025). RNNs and long short-term memory networks effectively process sequential data, such as analyzing language patterns in cover letters or interview dialogues to assess candidates' communication skills and emotional attitudes (Qin et al., 2020). CNNs, on the other hand, are applied to extract key features from resumes, helping recruitment systems automatically evaluate candidates' suitability for job requirements (Futia & Vetrò, 2020). The advantage of deep learning algorithms lies in their ability to extract deep features from raw data automatically, reducing the need for manual feature engineering. However, they often require large amounts of training data and significant computational resources.

Reinforcement learning algorithms are also starting to find applications in recruitment, particularly in candidate recommendation and decision optimization. Reinforcement learning optimizes decision strategies by interacting with the environment and receiving rewards or penalties (Ore & Sposato, 2022). In recruitment systems, reinforcement learning is used to optimize decision chains in the recruitment process, such as selecting the most suitable candidates or adjusting screening criteria to improve recruitment outcomes. By simulating the recruitment process, reinforcement learning can provide flexible, real-time decision support in complex recruitment scenarios, enhancing the precision and efficiency of hiring.

In addition to classification and evaluation algorithms, recommendation system algorithms have also received significant attention in recruitment applications. Recommendation systems analyze candidates' historical behavior, resume characteristics, and job requirements to recommend the most suitable positions to candidates or the most appropriate candidates to employers (Köchling et al., 2021; Xiao et al., 2020). Collaborative filtering, matrix factorization, and deep learning-based recommendation algorithms can build models based on historical data to predict the jobs candidates may be interested in and recommend the most suitable talent to employers (Köchling & Wehner, 2020; Pessach et al., 2020). These algorithms can process large-scale datasets and continuously optimize recommendations based on real-time changes in data, improving the precision and personalization of the recruitment process.

Overall, the application of algorithm optimization in recruitment has made recruitment systems smarter and more efficient. From resume screening to job recommendations, various algorithms enhance the speed and accuracy of recruitment through more efficient data processing and decision support, driving automation and personalization in recruitment processes.

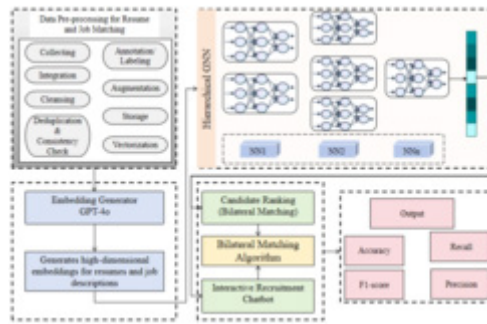
METHODS

System Architecture Overview

The intelligent talent management model proposed in this study consists of multiple modules designed to optimize different stages of the recruitment process. By integrating NLP and GNN, as well as matching algorithms, the system provides a comprehensive solution for recruitment, from data input to final candidate recommendations. Data preprocessing, feature extraction, candidate matching, and recommendation generation are included. On the input side, the system receives multisource data, such as candidates' resumes, job descriptions, interview feedback, and even social media information. They are preprocessed and converted into structured data for further analysis. During the processing phase, the system first utilizes the pretrained language model GPT-4o to extract features from textual data, capturing important information, such as hard skills, soft skills, work experience, and job

requirements. Afterward, a hierarchical GNN is used to model the multilevel relationships between candidates, job positions, and companies, constructing a multilevel graph structure. This graph maps connections between candidates and roles, optimizing the matching process. It also takes into account both individual and organizational needs. Building on this structure, the bilateral matching algorithm makes the final matching recommendation based on the mutual requirements of candidates and positions, and it continuously adjusts the matching strategy according to real-time feedback, thereby improving the accuracy and efficiency of the recruitment system. Through the close collaboration of these modules, the system can provide accurate candidate recommendations that meet the needs of all parties involved in the recruitment process. Figure 1 shows the overall system architecture and illustrates how the various modules work together.

Figure 1. Overall framework of the proposed recruitment matching model



Note. GPT-4o = generative pretrained transformer-4 omni, Hierarchical GNN = hierarchical graph neural network, NN = neural network.

Application of Pretrained Language Models in Resume and Job Matching

In this study, we used pretrained language models, particularly GPT-4o, as the core module for processing recruitment-related textual data. GPT-4o, a powerful NLP tool that can extract deep semantic features from unstructured textual data, such as resumes, cover letters, and job descriptions, can automatically assess whether a candidate meets the job requirements. Specifically, this process mainly involves text understanding, feature extraction, similarity computation, and recommendation generation. First and foremost, GPT-4o is used to process both the resumes and the job descriptions. A resume typically contains a candidate's personal information, educational background, work experience, skills, and expertise. On the other hand, job descriptions outline the responsibilities, qualifications, and required skills for a position. We inputted the resume and job description into GPT-4o, which then used its strong contextual understanding capabilities to extract key information from both. Building upon this analysis, GPT-4o generates recommendations regarding the suitability of a candidate for a specific role.

First, we needed to input each resume and job description into GPT-4o, which encodes them into high-dimensional semantic embeddings. These embeddings capture the textual information's key features, including hard skills, soft skills, work experience, and education. Let r_{resume} and r_{job} represent the embeddings for the resume and job description, respectively. These embeddings are generated using GPT-4o's encoding function shown in equation (1).

$$r_{\text{resume}} = f_{\text{GPT-4o}}(t_{\text{resume}}), r_{\text{job}} = f_{\text{GPT-4o}}(t_{\text{job}}) \quad (1)$$

In equation (1), $f_{\text{GPT-4o}}$ is the model's encoding function; t_{resume} and t_{job} are the raw textual inputs of the resume and job description, respectively; and r_{resume} , r_{job} are the resulting embeddings. Once the

embeddings are obtained, the initial similarity between the resume and job description is computed using the Euclidean distance, which measures the geometric distance between the two embedding vectors, as shown in equation (2).

$$d_{\text{euclidean}} = \| \mathbf{r}_{\text{resume}} - \mathbf{r}_{\text{job}} \| \quad (2)$$

In equation (2), $d_{\text{euclidean}}$ represents the dissimilarity between the resume and the job description based on their embeddings. Smaller values of $d_{\text{euclidean}}$ indicate greater similarity.

To better capture the relevance between specific features (e.g., skills, experience, education), a weighted cosine similarity score is introduced. This method accounts for the importance of individual features by assigning dynamic weights, as shown in equation (3).

$$S_{\text{cosine}} = \sum_{i=1}^n w_i \cdot \frac{\mathbf{r}_{\text{resume}}^i \cdot \mathbf{r}_{\text{job}}^i}{\| \mathbf{r}_{\text{resume}}^i \| \| \mathbf{r}_{\text{job}}^i \|} \quad (3)$$

In equation (3), w_i is the weight assigned to the i -th feature (e.g., skills or experience), $\mathbf{r}_{\text{resume}}^i$ and $\mathbf{r}_{\text{job}}^i$ are the corresponding feature embeddings, and S_{cosine} represents the weighted cosine similarity score. Additionally, GPT-4o's attention mechanism allows it to focus on the most relevant parts of resumes and job descriptions. A contextual interaction score is calculated using multi-head attention, which captures the alignment between tokens in the resume and job description, as shown in equation (4).

$$A_{\text{contextual}} = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \cdot V \quad (4)$$

In equation (4), Q , K , and V are the query, key, and value matrices derived from the embeddings, d is the embedding dimension, and $A_{\text{contextual}}$ represents the token-wise attention alignment between the two texts.

Finally, the overall compatibility score between a resume and a job is computed by combining multiple metrics: weighted cosine similarity, contextual interaction, and the Euclidean distance score. This combination ensures a comprehensive evaluation of candidate-job compatibility, as shown in equation (5).

$$R_{\text{total}} = \lambda_1 \cdot S_{\text{cosine}} + \lambda_2 \cdot (1 - d_{\text{euclidean}}) + \lambda_3 \cdot \text{mean}(A_{\text{contextual}}) \quad (5)$$

In equation (5), λ_1 , λ_2 , and λ_3 are hyperparameters controlling the contribution of each component, and R_{total} is the final matching score used to rank candidates for specific job positions.

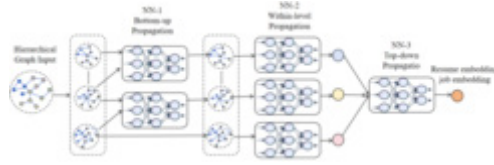
Through the methodology we just described, GPT-4o enables the extraction of both explicit and implicit features from recruitment data, such as skills, education, and experience, while also identifying contextual relationships that traditional methods might overlook. We designed the model to automatically prevent biases from being introduced during the matching process, ensuring fairness by excluding irrelevant demographic factors such as gender, age, or ethnicity. By automating the matching process, the model reduces manual effort, minimizes bias, and enhances matching accuracy. This approach allows the recruitment system to generate intelligent, personalized recommendations that significantly improve decision-making for employers and job seekers alike.

Hierarchical GNN

Hierarchical GNNs are developed to model the multilayered relationships among candidates, job positions, and companies. Although pretrained language models focus on text-based semantic

understanding, a hierarchical GNN leverages graph structures to analyze the interdependencies and indirect relationships in recruitment data. This approach provides a more comprehensive understanding of both direct and indirect connections, such as skill matching, alignment with company culture, and organizational compatibility. To ensure fairness in the matching process, a hierarchical GNN considers only relevant, nonbiased features, such as skills, experience, and job requirements, excluding demographic factors like gender, age, and ethnicity from the graph's structural analysis. As illustrated in Figure 2, in an RNN know as a bidirectional gated recurrent unit, the hierarchical GNN processes the input graph through three propagation layers (bottom-up, within-level, and top-down) to generate high-dimensional embeddings for candidates and job positions. These embeddings are subsequently used by the bilateral matching algorithm to calculate the final matching score.

Figure 2. The structure of a bidirectional gated recurrent unit



Note. NN = neural network.

The following formulas describe the gate update mechanism, candidate hidden state calculation, final hidden state calculation, and output calculation in the bidirectional gated recurrent unit model in detail.

The recruitment system is represented as a graph $G = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes the nodes (candidates, jobs, companies) and $\mathcal{E}$ represents the edges that define relationships between the nodes. Each node $v \in \mathcal{V}$ is associated with an initial feature vector $h_v^{(0)}$, which integrates information from textual embeddings (e.g., GPT-4o) and graph-specific features, such as organizational structure or hiring priorities. To update the node representations, a hierarchical GNN employs a message-passing mechanism. At each layer l , a node v 's representation is updated based on its own features and those of its neighbors $u \in \mathcal{N}(v)$. The key update rule is calculated as shown in equation (6).

$$h_v^{(l)} = \sigma \left(W_{\text{self}}^{(l)} h_v^{(l-1)} + \sum_{u \in \mathcal{N}(v)} \frac{1}{\sqrt{|\mathcal{N}(v)|}} W_{\text{neighbor}}^{(l)} h_u^{(l-1)} e_{uv} \right) \quad (6)$$

In equation (6), $W_{\text{self}}^{(l)}$ and $W_{\text{neighbor}}^{(l)}$ are trainable weight matrices for self-features and neighbor features, respectively; $\mathcal{N}(v)$ is the set of neighbors of v ; e_{uv} is the edge weight between u and v ; and σ is an activation function, such as rectified linear unit.

To capture the relationship strength between nodes, the edge weight e_{uv} is calculated dynamically as shown in equation (7).

$$e_{uv} = \text{sigmoid}(h_u^{(0)} \cdot h_v^{(0)} + b_{uv}) \quad (7)$$

In equation (7), b_{uv} is a bias term that captures additional contextual factors, such as the frequency of interaction between the nodes or the relevance of the connection type.

As the graph is processed across multiple layers, the final representation of each node aggregates the information from each layer using weighted sums. This calculation is shown in equation (8).

$$h_v^{agg} = \sum_{l=1}^L \alpha_l h_v^{(l)} \quad (8)$$

In equation (8), L is the total number of layers, and α_l is a trainable parameter controlling the contribution of each layer to the final representation.

Once the node representations are computed, the compatibility score between a candidate c and a job j is calculated as shown in equation (9).

$$S_{match}(c,j) = h_c^{agg} \cdot h_j^{agg} + \beta \cdot g(c,j) \quad (9)$$

In equation (9), h_c^{agg} and h_j^{agg} are the aggregated representations of the candidate and job nodes, $g(c,j)$ is a graph-based scoring function that captures higher-order interactions (e.g., cultural fit or career trajectory alignment), and β is a hyperparameter controlling the contribution of the graph-based score.

To ensure that the representations are consistent with the graph's structural relationships, a graph regularization term is added during training. This calculation is shown in equation (10).

$$\mathcal{L}_{reg} = \sum_{(u,v) \in \mathcal{E}} \|h_u^{agg} - h_v^{agg}\|^2 \cdot e_{uv} \quad (10)$$

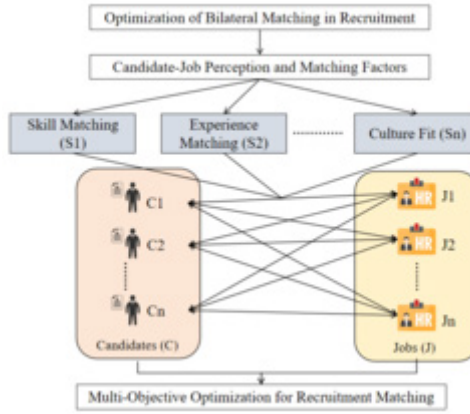
This regularization term penalizes large differences between connected nodes based on their edge weights, encouraging similar nodes to have similar representations.

Hierarchical GNN effectively models the complexities of recruitment data by integrating textual features, structural dependencies, and multilayered relationships. It enables the recruitment system to provide more precise, context-aware recommendations, ensuring that the matching process accounts for both direct candidate-job compatibility and broader organizational alignment. By eliminating bias during the graph construction and processing stages, the system ensures that recommendations are not influenced by irrelevant demographic factors, leading to fairer and more equitable matching decisions.

Bilateral Matching Algorithm

We designed the bilateral matching algorithm to account for the bidirectional nature of candidate-job relationships. Unlike conventional one-sided matching systems, this approach simultaneously considers the requirements and preferences of both candidates and job positions. By integrating multiple matching factors, such as skill alignment, experience matching, and cultural fit, as well as incorporating personalized weights, historical data, and real-time adjustments, the algorithm ensures a balanced and efficient matching process. Figure 3 illustrates the structure of the bilateral matching algorithm, showcasing how candidates (C) and job positions (J) are interconnected through multi-objective optimization based on these key matching factors.

Figure 3. Structure of the bilateral matching algorithm



Note. HR = human resources.

To begin the design for our model, we define the compatibility between a candidate c and a job j based on their respective feature representations. The preliminary compatibility score is calculated as a combination of candidate-specific features h_c and job-specific features h_j , as shown in equation (11).

$$S_{\text{preliminary}}(c,j) = h_c \cdot h_j \quad (11)$$

In equation (11), h_c and h_j are the feature vectors of the candidate and job, respectively. This dot product captures the overall similarity of their features.

Next, to account for individual preferences and job priorities, the algorithm introduces bias terms for both candidates and jobs. These terms allow the system to adjust for unique factors, such as a candidate's preferred job location or a company's priority for specific skills, as shown in equation (12).

$$S_{\text{personalized}}(c,j) = S_{\text{preliminary}}(c,j) + b_c + b_j \quad (12)$$

In equation (12), b_c is the candidate bias term and b_j is the job bias term. These biases reflect external preferences or trends not directly encoded in the feature vectors. The algorithm also integrates historical recruitment data to dynamically adjust the matching score. Let D_{history} represent the historical dataset, which includes previous successful matches, candidate performance metrics, and job-specific hiring trends. A correction factor is computed from D_{history} , and the matching score is updated accordingly, as shown in equation (13).

$$S_{\text{adjusted}}(c,j) = S_{\text{personalized}}(c,j) \cdot (1 + \beta \cdot f(D_{\text{history}})) \quad (13)$$

In equation (13), $f(D_{\text{history}})$ is a function that models the impact of historical data on the current match, and β is a hyperparameter controlling the weight of historical adjustments. This adjustment factor ensures that past hiring trends influence the matching process in a meaningful way.

To further incorporate real-time feedback from recruiters or candidates, the algorithm recalibrates the scores based on recent feedback. This calculation is shown in equation (14).

$$S_{\text{real-time}}(c,j) = S_{\text{adjusted}}(c,j) + \gamma \cdot \Delta(c,j) \quad (14)$$

In equation (14), $\Delta(c, j)$ represents real-time feedback adjustments based on recruiter evaluations or candidate actions (e.g., applying for a position, rejecting an offer), and γ is a hyperparameter that controls the sensitivity to real-time updates. Finally, the overall compatibility score is normalized and aggregated across all candidate-job pairs. The algorithm generates a ranked list of matches, optimizing for both candidate satisfaction and job-specific priorities, as shown in equation (15).

$$M_{\text{ranked}} = \text{Rank}(\{S_{\text{real-time}}(c, j) \mid c \in \mathcal{C}, j \in \mathcal{J}\}) \quad (15)$$

In equation (15), $\mathcal{C}$ is the set of all candidates, $\mathcal{J}$ is the set of all job positions, and $\text{Rank}(\cdot)$ denotes the ranking function that orders candidate-job pairs by their final compatibility scores.

By combining static feature compatibility, historical data insights, and real-time feedback, the algorithm enhances the accuracy, adaptability, and fairness of the recruitment system. It enables the recruitment system to provide more precise, context-aware recommendations, ensuring that the matching process accounts for both direct candidate-job compatibility and broader organizational alignment. By eliminating bias during the graph construction and processing stages, the system ensures that recommendations are not influenced by irrelevant demographic factors, leading to fairer and more equitable matching decisions.

EXPERIMENT

Datasets

The Resume Corpus Dataset is a publicly available dataset focused on resume parsing. It includes a large collection of anonymized real-world resumes spanning various industries such as information technology, finance, education, and health care. Each record in the dataset is a complete resume text, providing key information about the candidate, such as skills, educational background, work experience, project history, and professional domains.

The dataset exhibits high diversity, reflecting the varying characteristics of candidates in real recruitment scenarios. For example, a sample resume may include a statement such as, “John Doe is a software engineer with 5 years of experience in Python, Java, and cloud computing. He holds a master’s degree in computer science.” This dataset is primarily used for candidate feature extraction tasks, such as skill identification, work experience classification, and educational level analysis, forming the basis for generating unified embeddings for the model.

The curriculum vitae-job description (CV-JD) Matching Dataset is specifically designed for resume-job matching tasks, containing numerous pairs of resumes and job descriptions with associated labels or scores indicating their compatibility. Each record consists of a resume text, a job description text, and a matching relationship label (e.g., “match” or “no match”). Job description texts typically outline responsibilities, required skills, experience levels, and other hiring criteria. For example, a description may be as follows: “We are seeking a software developer with 3+ years of experience in Python and Java, with a background in cloud-based system design.” To further illustrate the characteristics of job descriptions in the dataset, Figure 4 presents a keyword cloud map generated from the most frequent terms in job descriptions. The most prominent keywords include “data analysis,” “Python,” “communication skills,” and “bachelor’s degree or above,” reflecting common requirements across industries and the dataset’s relevance to real-world recruitment scenarios.

Figure 4. Job requirement keyword cloud map



These two datasets complement each other in functionality. The Resume Corpus Dataset provides rich multidimensional candidate features, making it ideal for generating feature representations for the model. The CV-JD Matching Dataset, on the other hand, offers supervised training samples for resume-job matching tasks, enabling the model to be trained and evaluated on realistic matching scenarios. Combining these two datasets enables our model’s performance to be comprehensively evaluated in both feature extraction and matching tasks, demonstrating its potential value in real-world recruitment systems.

Data Preprocessing

To ensure high data quality and compatibility with the proposed model, both the Resume Corpus Dataset and CV-JD Matching Dataset were subjected to a series of preprocessing steps, including text cleaning, data splitting, and normalization.

At the very beginning, we cleaned all textual data to remove unnecessary elements, such as stop words, punctuation, and special characters. We also converted the text to lowercase to standardize formatting and reduce inconsistencies caused by capitalization. For example, the original resume text “John is a Software Engineer with 5+ years of experience in Python, Java, and Cloud Computing,” was processed into “john software engineer 5 years experience python java cloud computing.” These steps ensured the text was in a standardized format for subsequent processing. Following the text cleaning, we generated semantic embeddings for resumes and job descriptions using the pretrained GPT-4o model. As described in equation (1) earlier, these embeddings provide high-dimensional semantic representations of resumes (r_{resume}) and job descriptions (r_{job}). These embeddings are then used in the subsequent hierarchical graph representation and bilateral matching process, providing a unified feature format for the matching tasks. After the embedding generation, we split the datasets into training, validation, and test sets to ensure unbiased evaluation of the model. The datasets were divided using a ratio of 70% for training, 15% for validation, and 15% for testing. During the split, we carefully maintained the distribution of matching labels in the CV-JD Matching Dataset across the subsets to prevent label imbalance that could bias the evaluation.

To improve model stability, normalization was applied to the embedding vectors. Each feature of the embeddings was standardized to have zero mean and unit variance using the formula shown in equation (16).

$$r_{\text{normalized}} = \frac{r - \mu}{\sigma} \quad (16)$$

In equation (16), μ is the mean of the embedding features, and σ is the standard deviation. This normalization step ensured consistent magnitudes of the features and avoided numerical instability during training.

For the CV-JD Matching Dataset, if the matching labels were provided as continuous scores (e.g., ranging from 0 to 1), we applied a threshold value (e.g., 0.5) to binarize the labels into 0 (no match) and 1 (match). This transformation simplified the matching task into a binary classification problem and made it easier to evaluate the performance of the proposed model. Lastly, we implemented quality control measures to filter out noisy or problematic samples. For instance, resumes and job descriptions that were excessively short or long were removed, ensuring uniformity in the dataset. Additionally, we examined the label distribution across the training, validation, and test sets to ensure consistency and avoid imbalance that could impact the evaluation.

Through these preprocessing steps, the Resume Corpus Dataset was transformed into structured feature representations for candidates, while the CV-JD Matching Dataset was prepared for supervised evaluation of resume-job compatibility. These steps ensured that the input data met the requirements of the proposed model and established a reliable foundation for subsequent experiments.

Experimental Setup

The experiments were conducted on a high-performance computing server equipped with an Nvidia A100 graphics processing unit, Intel Xeon processor, and 128 GB of memory. The server ran Ubuntu 20.04, and the main development frameworks used were PyTorch 2.0 and Hugging Face Transformers. Throughout the experiments, we maintained a consistent environment to ensure the reproducibility of results.

During the training phase, the pretrained language model GPT-4o was employed to generate semantic embeddings for resumes and job descriptions. The hierarchical GNN and bilateral matching algorithm were adopted for multilevel feature relationship modeling and matching score computation. In addition, the Adam optimizer was used with an initial learning rate of 10^{-4} , and a linear learning rate decay strategy was applied to gradually reduce the learning rate during training. A batch size of 32 was used, and the model underwent training for a total of 50 epochs. An early stopping strategy was applied to prevent overfitting, where training was automatically stopped if no significant improvement in the validation set performance was observed for five consecutive epochs. Additionally, a dropout rate of 0.3 was applied to the hierarchical GNN layers, and L2 regularization with a weight decay factor of 10^{-5} was introduced to further enhance the model's generalization ability.

The dimensionality of the embeddings was set to 1,024. The high-dimensional embeddings generated by GPT-4o were processed through the hierarchical GNN for multilevel relationship modeling, and the final node representation dimensionality was reduced to 256. The model hyperparameters were fine-tuned throughout the training process to achieve optimal performance on the validation set.

After preprocessing, the data were divided into training, validation, and test sets with a ratio of 70%, 15%, and 15%, respectively. The performance on the validation set was continuously monitored during training to ensure dynamic adjustment of hyperparameters and the stability of the training process.

Experimental Metrics

We evaluated the performance of the proposed model using standard metrics widely employed in classification tasks. Accuracy was used to measure the overall correctness of the predictions, while precision and recall provided insights into the model's ability to correctly identify positive matches and capture all relevant matches, respectively. To balance precision and recall, the F1 score was adopted as a

comprehensive metric to assess classification performance. In addition, area under the curve (AUC) and receiver operating characteristic measured the model’s ability to distinguish between matched and non-matched samples. Therefore, it provided an overall indicator of classification quality. Overall, these metrics were chosen to ensure a robust and holistic evaluation of the model’s effectiveness in matching resumes to job descriptions.

Experimental Results and Analysis

To evaluate the performance of the proposed model, we compared it against several baseline and state-of-the-art models commonly used for feature extraction in resume data (Table 1). These models included traditional embedding techniques, like Global Vectors for Word Representation (GloVe) and Word2Vec, as well as more advanced pretrained language models, such as Bidirectional Encoder Representations from Encoders (BERT), DistilBERT, and RoBERTa. Although traditional models like GloVe and Word2Vec rely on static word embeddings, newer transformer-based models like BERT and RoBERTa leverage contextual understanding of language, thus allowing them to capture richer semantic relationships.

Table 1. Comparative results on the resume corpus dataset for feature extraction

Model	Accuracy	Precision	Recall	F1 Score
GloVe	0.74	0.72	0.7	0.71
Word2Vec	0.78	0.76	0.73	0.74
BERT	0.85	0.83	0.8	0.81
DistilBERT	0.83	0.81	0.79	0.80
RoBERTa	0.87	0.85	0.83	0.84
Proposed model	0.89	0.87	0.85	0.86

Note. BERT = Bidirectional Encoder Representations from Encoders, GloVe = Global Vectors for Word Representation.

The results in Table 1 compare the performance of the proposed model with baseline and state-of-the-art models on the Resume Corpus Dataset for feature extraction. The proposed model achieves the highest scores across all evaluation metrics: accuracy of 0.89, precision of 0.87, recall of 0.85, and an F1 score of 0.86. These results demonstrate that the integration of GPT-4o embeddings and the hierarchical GNN effectively captures the intricate semantic features within resumes, which means superior performance in feature extraction. In contrast, traditional embedding models like GloVe and Word2Vec attain lower F1 scores of 0.71 and 0.74, respectively, owing to their static natures. This lower performance limits their ability to represent contextual information. Pretrained language models such as BERT and DistilBERT show improved performance with F1 scores of 0.81 and 0.80, respectively, owing to their ability to capture richer contextual relationships. RoBERTa achieves a competitive F1 score of 0.84. This figure reflects its strength as an advanced transformer-based model. However, the proposed model outperforms RoBERTa by incorporating hierarchical graph structures through the hierarchical GNN. This approach allows for multilevel feature aggregation and better modeling of the relationships within the resume data. This hierarchical approach enables the model to capture not just the semantic content but also the structural nuances of resumes, leading to more comprehensive feature representations. The analysis indicates that combining advanced pretrained embeddings from GPT-4o with hierarchical GNN provides a significant advantage over both static embedding methods and other transformer-based models. The superior performance of the proposed

model validates its effectiveness in extracting high-quality semantic features from resumes, which is crucial for enhancing downstream tasks such as resume-job matching and candidate recommendation.

The results in Table 2 illustrate the performance comparison of the proposed model against other baseline and advanced models on the CV-JD Matching Dataset for the job-resume matching task. The proposed model achieves the highest scores across all evaluation metrics: accuracy of 0.92, precision of 0.91, recall of 0.90, and an F1 score of 0.90. These results highlight the model's effectiveness and robustness in capturing both semantic and structural relationships between resumes and job descriptions.

Table 2. Comparative results on the curriculum vitae-job description matching dataset for the matching task

Model	Accuracy	Precision	Recall	F1 Score
Cosine similarity-based matching (Rinjeni et al., 2024)	0.74	0.72	0.69	0.70
NSEEN (Fakhraei et al., 2020)	0.80	0.78	0.75	0.76
SSMI (Qi & Qin, 2023)	0.84	0.82	0.8	0.81
AdamGNN (Zhong et al., 2022)	0.88	0.86	0.84	0.85
SuperGlue (Sarlin et al., 2020)	0.90	0.89	0.87	0.88
Proposed model	0.92	0.91	0.90	0.90

Note. NSEEN = Neural Semantic Embedding for Entity Normalization, SSMI = Semantic Similarity and Mutual Information Maximization.

Specifically, the baseline model, cosine similarity-based matching, performs the worst among all compared models, with an F1 score of 0.70, reflecting its limitation in modeling complex semantic relationships. This poor performance of the baseline model is because it relies solely on simple similarity calculations between embeddings. As for the Neural Semantic Embedding for Entity Normalization (NSEEN) model, it performs slightly better, with an F1 score of 0.76. This model benefits from enhanced semantic features, but it still lacks the ability to model intricate relationships. The Semantic Similarity and Mutual Information Maximization (SSMI) model further improves to an F1 score of 0.81 by incorporating better semantic matching mechanisms, showing the importance of refined semantic understanding in matching tasks. More advanced models, such as AdamGNN and SuperGlue, achieve significantly better results, with F1 scores of 0.85 and 0.88, respectively. AdamGNN benefits from its GNN framework, which captures the interactions between resumes and job descriptions more effectively by modeling candidate-job relationships as a graph structure. SuperGlue, an even more advanced model, integrates multiple layers of feature refinement and matching mechanisms, achieving near-optimal results among the baselines.

The proposed model outperforms even the advanced SuperGlue model by achieving an F1 score of 0.90. This improvement highlights the importance of integrating hierarchical graph modeling with bilateral matching algorithms; this enhancement allows the model to capture bidirectional preferences and semantic relationships more comprehensively. The proposed model's ability to surpass both baseline and advanced methods demonstrates its effectiveness in handling the challenges of resume-job matching tasks, including complex feature relationships, semantic nuances, and the need for accurate compatibility scoring. This result validates the design and implementation of the proposed approach as a state-of-the-art method for matching tasks in recruitment systems.

To evaluate the models comprehensively, we present in Table 3 the accuracy, F1 score, and inference time of each model on the CV-JD matching task across small, medium, and large datasets. This comparison helps to highlight each model's performance and efficiency at different data scales.

Table 3. Performance comparison on different data scales for the curriculum vitae-job description matching task

Model	Small (10k)		Medium (50k)		Large (200k)	
	Accuracy	F1 Score	Accuracy	F1 Score	Accuracy	F1 Score
Cosine similarity-based matching	0.74	0.7	0.76	0.73	0.77	0.74
NSEEN	0.80	0.76	0.81	0.78	0.82	0.79
SSMI	0.84	0.81	0.85	0.82	0.86	0.83
AdamGNN	0.88	0.85	0.89	0.86	0.90	0.87
SuperGlue	0.90	0.88	0.91	0.89	0.92	0.90
Proposed model	0.92	0.90	0.93	0.91	0.94	0.92

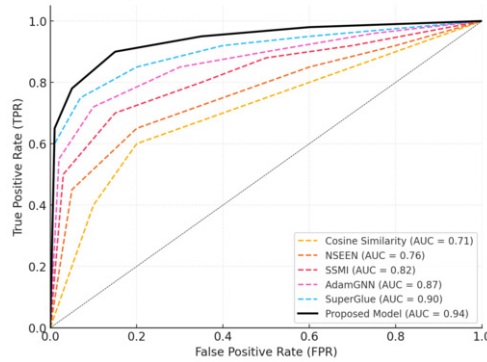
Note. NSEEN = Neural Semantic Embedding for Entity Normalization, SSMI = Semantic Similarity and Mutual Information Maximization.

Table 3 demonstrates the performance of the models on the CV-JD matching task across different data scales. As shown, the proposed model consistently outperforms all other models, with significant improvements as the dataset size increases. In the large dataset (200k), the proposed model achieves an accuracy of 0.94 and an F1 score of 0.92, far surpassing the other models. The cosine similarity-based matching model exhibits weaker performance, with significantly lower accuracy and F1 score compared with the other methods. NSEEN and SSMI show solid performances on smaller datasets, but lose their advantages as the data size grows. AdamGNN and SuperGlue show stable improvement as the dataset size increases, with SuperGlue performing nearly as well as the proposed model on large datasets.

In terms of computational efficiency, the proposed model does have higher inference time compared with simpler models, such as cosine similarity-based matching, but it remains competitive with more advanced models, like SuperGlue and AdamGNN. Despite its higher computational cost, the proposed model provides the best trade-off between accuracy and computational efficiency, especially when scaling up to larger datasets. This finding shows that the proposed model, combining GPT-4o embeddings and hierarchical GNN, not only leads on smaller datasets but also maintains excellent performance on large datasets, demonstrating superior stability and adaptability in real-world recruitment tasks.

Figure 5 shows the receiver operating characteristic curves for models evaluated on the CV-JD Matching Dataset. The curves highlight the trade-off between the true positive rate and false positive rate at various thresholds. The AUC quantifies each model’s classification effectiveness.

Figure 5. The comparison of different models in different indicators on the datasets

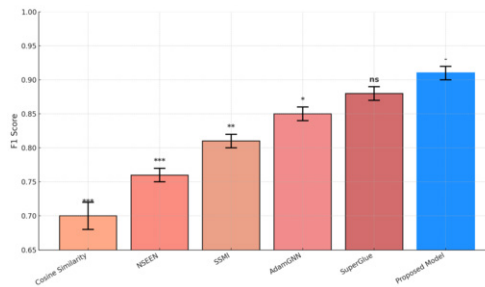


Note. AUC = area under the curve, NSEEN = Neural Semantic Embedding for Entity Normalization, SSMI = Semantic Similarity and Mutual Information Maximization.

The proposed model achieves the highest AUC of 0.94, demonstrating its exceptional ability to distinguish between matched and unmatched pairs. Its steep curve indicates high true positive rates even at low false positive rates, which highlights the superior performance resulting from the integration of GPT-4o embeddings, hierarchical GNN, and bilateral matching algorithm. The SuperGlue model follows with an AUC of 0.90, indicating strong semantic matching performance, although slightly less sensitive than the proposed model. AdamGNN, with an AUC of 0.87, reflects solid graph-based modeling, but lacks the fine-grained semantic understanding of the proposed model. SSMI, achieving an AUC of 0.82, performs reasonably well, but is limited by its inability to capture hierarchical relationships. The NSEEN model, with an AUC of 0.76, shows modest improvement over simple methods, but struggles with complex matching. Finally, cosine similarity-based matching, with an AUC of 0.71, demonstrates the weakest performance owing to its reliance on static similarity measures. These results underscore the critical contributions of hierarchical graph modeling and bilateral matching in achieving state-of-the-art performance. The proposed model outperforms all other methods by combining advanced semantic embeddings with robust structural relationship modeling, as demonstrated by its superior AUC score.

Figure 6 presents the F1 score (mean $\pm$ standard deviation) for various models across multiple experimental runs. Statistical significance levels are annotated above each bar: “” for $p < 0.001$, “” for $p < 0.01$, “” for $p < 0.05$, and “ns” for not significant. The proposed model is highlighted in blue, showcasing its superior performance.

Figure 6. F1 score comparison across different models with statistical significance analysis



Note. NSEEN = Neural Semantic Embedding for Entity Normalization, SSMI = Semantic Similarity and Mutual Information Maximization.

Figure 6 illustrates the F1 score for each model and highlights the statistical significance of the differences between the proposed model and the other models. The proposed model achieves the highest F1 score, with a mean of 0.91 ± 0.01 , so it outperforms all baseline and advanced models. Its low standard deviation indicates strong robustness and consistency across multiple experimental runs. The cosine similarity model performs the weakest, with an F1 score of 0.70 ± 0.02 , and the difference compared with the proposed model is statistically significant ($p < 0.001$). Similarly, NSEEN and SSMI perform better than cosine similarity, achieving F1 scores of 0.76 ± 0.01 and 0.81 ± 0.01 , respectively, but both are significantly outperformed by the proposed model ($p < 0.001$ and $p < 0.01$, respectively). Among more advanced models, AdamGNN achieves an F1 score of 0.85 ± 0.01 , showing robust performance, but the proposed model still shows a statistically significant improvement ($p < 0.05$). The SuperGlue model performs competitively with an F1 score of 0.88 ± 0.01 , and the p-value ($p = 0.07$) indicates that its difference from the proposed model is not statistically significant, highlighting its close performance to the proposed model. Overall, the results validate the effectiveness of the Proposed Model, which outperforms baseline and state-of-the-art models in terms of both F1 score and robustness. The statistical significance analysis reinforces the claim that the proposed model offers meaningful improvements, particularly when compared with simpler or traditional approaches. The consistent performance across multiple runs further demonstrates its capability to handle the complex task of resume-job matching effectively.

The ablation study results presented in Table 4 demonstrate the importance of each key component of the proposed model. The full model achieves the best performance across all metrics, with an accuracy of 0.92 and an F1 score of 0.90. These results showcase the synergistic effect of combining GPT-4o embeddings, hierarchical GNN, and the bilateral matching algorithm.

Table 4. Ablation study results on the curriculum vitae-job description matching dataset

Ablation Setting	Accuracy	Precision	Recall	F1 Score
Full model (proposed)	0.92	0.91	0.90	0.90
Without GPT-4o embeddings	0.88	0.86	0.85	0.85
Without hierarchical GNN	0.87	0.85	0.84	0.84
Without bilateral matching algorithm	0.89	0.87	0.86	0.86

continued on following page

Table 4. Continued

Ablation Setting	Accuracy	Precision	Recall	F1 Score
Without both GPT-4o and GNN	0.84	0.82	0.80	0.81
Without both GNN and bilateral matching	0.85	0.83	0.81	0.82

Note. GNN = graph neural network.

Replacing GPT-4o embeddings with simpler embeddings (e.g., BERT) results in an immense drop in performance, particularly in precision and recall, with the F1 score decreasing to 0.85. This result emphasizes the critical role of advanced pre-trained embeddings in accurately capturing semantic information. When the hierarchical GNN is removed, the model's ability to capture multilevel structural relationships between resumes and job descriptions is also compromised. In this context, there is a decline in all metrics, with the F1 score falling to 0.84. This finding underscores the critical role of the GNN in leveraging graph-based relationships to improve the matching process. Similarly, removing the bilateral matching algorithm leads to a notable drop in recall, indicating that this component is instrumental in capturing bidirectional compatibility between resumes and job descriptions.

The ablation study also explored the impact of simultaneously removing multiple components. When both GPT-4o embeddings and the GNN are excluded, the F1 score drops further to 0.81, illustrating the complementary nature of these components. Additionally, removing both the GNN and the bilateral matching algorithm reduces the F1 score to 0.82, emphasizing the necessity of these modules working together to ensure optimal performance. These results validate the design choices of the proposed model and demonstrate that each component contributes significantly to the overall effectiveness of the system.

LIMITATIONS AND FUTURE WORK

Although the proposed model demonstrates strong performance on benchmark datasets, it still faces certain limitations that constrain its immediate deployment in practical recruitment environments. One major limitation is its reliance on large-scale pretrained language models such as GPT-4o, which demands substantial computational resources and may not be easily adopted by smaller organizations or low-resource settings. Additionally, the hierarchical graph structure, although effective in modeling complex relationships, can be sensitive to noisy or highly heterogeneous data, thus requiring careful feature engineering and parameter tuning. Furthermore, the datasets used in this study are publicly available and, despite their diversity, cannot fully capture the nuanced dynamics of real-world corporate recruitment, such as evolving job roles, domain-specific vocabulary, and implicit organizational preferences. In future work, we plan to incorporate real recruitment data from enterprise contexts to better reflect authentic hiring conditions and improve the practical applicability of the model. Moreover, we aim to explore lightweight model architectures or knowledge distillation techniques to reduce deployment costs. Finally, expanding the model's capabilities to include soft skill analysis, cultural fit evaluation, and long-term employee potential will help create a more holistic and robust intelligent recruitment system that aligns with real hiring decisions.

CONCLUSION

In this study, we proposed a novel resume-job matching model, combining GPT-4o embeddings, hierarchical GNN, and a bilateral matching algorithm to effectively capture semantic, structural, and bidirectional relationships. Experimental results on publicly available datasets demonstrate that our model outperforms baseline and state-of-the-art approaches in terms of F1 score, accuracy, and

robustness. Statistical significance analysis further confirms the superiority of the proposed model. This model provides a more efficient and accurate approach to the resume-job matching task, thereby benefitting recruitment systems by automating and improving decision-making for employers and job seekers.

COMPETING INTERESTS

We declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential competing interest.

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